Mangrove structure and topography: field form

One form per transect. Print pages 1–2 (header and topography) for every transect, plus the pages of the structure method used: A (page 3), B (page 4) or C (pages 5–6); page 7 is the post-site report, filled in the same day. Fill it in with Adobe Acrobat Reader (free): the Transect ID is then filled in automatically and the form is saved correctly. Method: Mangroves and Field transects pages of the SEASTRONG Field Processing Tools documentation.

1. Transect header

Study area (BZ / SC / FJ) Belize, Seychelles or Fiji		**Corridor (C1–C6)** Corridor number in the study area	
Transect ID e.g. BZ160327C1mangrove (automatic in Acrobat)	will autocompile	**Date** Survey date, as mm/dd/yyyy	
Operators Names of the people in the field		**Structure method** A photogrammetry, B tree-by-tree, C manual	
Sensor 1 ID (seaward) Serial number of the seaward pressure sensor		**Sensor 1 position on tape (m)** Distance along the tape, e.g. 0	
Sensor 2 ID (landward) Serial number of the landward pressure sensor		**Sensor 2 position on tape (m)** Distance along the tape, e.g. 30	
GPS point at 0 m end (lat, lon) One approximate point at the 0 m end		**Bearing 0 → 30 m (°)** Compass bearing from 0 m towards 30 m	
Time zone of watch / phone Clock used for the times, e.g. UTC-6		**Low tide time** Time of low water on the survey day	
Dominant species e.g. Rhizophora mangle		**Forest description** Open or dense; pneumatophores or prop roots	

Methods along the tape If more than one, e.g. A 0–12 m, C 12–30 m	
Vegetation changes along tape (m: description) e.g. 0–6 m pneumatophores only; 6–30 m dense prop roots	
Vegetation beyond the 30 m mark Short description of the 5–10 m beyond the tape	

2. Before starting

- Working at low tide, forest floor exposed
- 30 m tape laid from sensor 1 (0 m, forest edge) to sensor 2 (30 m), straight and secured
- Overview photos of the transect taken from both ends
- Equipment for the topography: 50 m tape, clinometer, plastic board, station pegs, GNSS
- Equipment for the structure method (see the page of the method used)

3. Topography: from the low-water mark to 5–10 m beyond the 30 m mark

Low-water mark: time Time when the low-water mark was identified		**Low-water mark: position** Distance seaward of the 0 m mark (m)	
GNSS at start: lat, lon At the low-water mark		**GNSS at start: elevation (m)** RTK only; leave empty with handheld GNSS	
GNSS type RTK or handheld		**Datum tie at this site** RTK fix or benchmark, at least once per site	

Distance: cumulative distance along the taut tape from the low-water mark (slope distance, m). **Inclination**: slope of the segment leading to this station, clinometer on the plastic board (°; **+ uphill landward**, – downhill landward; empty on the first row). **Label**: low water, 0 m, sensor 1, sensor 2, 30 m, end. Stations at every change in slope and at most 2 m apart.

Station	Distance (m)	Inclination (°)	Label	Station	Distance (m)	Inclination (°)	Label
1				21			
2				22			
3				23			
4				24			
5				25			
6				26			
7				27			
8				28			
9				29			
10				30			
11				31			
12				32			
13				33			
14				34			
15				35			
16				36			
17				37			
18				38			
19				39			
20				40			

4. Method A: photogrammetry

Camera spacing (cm) Measured distance between the two lens centres		**Rig height above ground (m)** About 1–1.5 m	
Cameras start time Time both cameras were started		**Cameras end time** Time both cameras were stopped	

- [ ] Two cameras set to the same time (GoPro app); identical settings; white balance fixed; no red filter
- [ ] At least 6 scale bars on both sides of the tape, within ~2 m, varied orientation and elevation (used only for scaling)
- [ ] Pass 1: forward, camera A over the tape, cameras pointing down, ~0.3 m/s
- [ ] Pass 2: backward, camera A over the tape, cameras pointing down
- [ ] Pass 3: zigzag across the tape to ~3 m each side, cameras tilted ~45° forward
- [ ] Photos checked on both cameras: complete and sharp

Scale bars: target codes of the bar, distance along the tape (m), side (L or R looking landward).

Bar	Target codes	Position on tape (m)	Side (L/R)
e.g.	*T01 / T02*	*2.5*	*L*
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Passes repeated? (which, why)	

5. Method B: tree-by-tree (Mori et al., 2022)

Trees in front plot (n) All trees rooted in the 5 m × 5 m plot		**Trees in centre plot (n)** All trees rooted in the 5 m × 5 m plot	
Trees in back plot (n) All trees rooted in the 5 m × 5 m plot			

At least 5 sample trees per plot, covering the range of sizes. Heights and lengths in cm, diameters in cm. **H**: tree height · **CB**: crown-base height (lowest major branch) · **DBH**: at 1.3 m · **D1**: trunk just above the roots · **Hroot**: top of the root system · **W1–W6**: trunk to outermost root in six directions · **N**: prop roots reaching the ground · **d20**: mean root diameter at 20 cm · **Ph**: root photos taken in two directions (Y/N).

Plot	Tree	H	CB	DBH	D1	Hroot	W1	W2	W3	W4	W5	W6	N	d20	Ph
front	1														
front	2														
front	3														
front	4														
front	5														
centre	1														
centre	2														
centre	3														
centre	4														
centre	5														
back	1														
back	2														
back	3														
back	4														
back	5														

6. Method C: manual measurements

Density: in each 5 m × 5 m plot, count all trees and measure the DBH of each (at 1.3 m, or 30 cm above the highest prop root: write that height). **Pneumatophores**: one 1 m × 1 m quadrat per plot (0.5 m × 0.5 m if too dense: write the size); count all, and measure height and diameter of up to 20.

Plot	Trees (n)	Species	DBH of each tree (cm)	Quadrat size	Pneumat. (n)	Pneumat. heights / diameters (cm)
front						
centre						
back						

Sample trees (3–5 per plot): at each height, the number of prop roots of this tree crossing that height, their diameters (all, or up to 10), and the trunk diameter. Also record root-system height (Hroot), crown-base height and tree height. Print this page twice for more than 3 trees per plot.

Plot / tree	Hroot / CB / H (cm)	Height (cm)	Roots crossing (n)	Root diameters (cm)	Trunk diam. (cm)
front 1		25			
		50			
		75			
		100			
		150			
front 2		25			
		50			
		75			
		100			
		150			
front 3		25			
		50			
		75			
		100			
		150			
centre 1		25			
		50			
		75			
		100			
		150			
centre 2		25			
		50			
		75			
		100			
		150			
centre 3		25			
		50			
		75			
		100			
		150			
back 1		25			
		50			
		75			
		100			
		150			
back 2		25			
		50			
		75			
		100			
		150			
back 3		25			
		50			
		75			
		100			
		150			

Post-site report

Filled in the same day, after downloading the data. Send it with the data to the data manager.

A. Survey

Transect ID e.g. BZ160327C1mangrove (automatic in Acrobat)	will autocompile	**Date** Survey date, as mm/dd/yyyy	
Filled in by Who fills in this report		**Survey complete (Y / N)** Topography and the structure method completed	
Time on site (h) From arrival to departure		**Weather** Sun, cloud, rain	
Ground conditions e.g. soft mud, firm sand, standing water			

B. Data

Topography CSV prepared (Y / N) Stations typed into the CSV for the notebook		**Photos (n)** Total number of photos from the survey	
Photo folder name Transect ID + _photos		**Slate / sheets photographed (Y / N)** Photos of all paper sheets saved with the data	

Data download link Link to the shared folder with all data, photos and this form	

C. Quality check (tick if a problem)

- Tide came in before the survey was finished
- Topography incomplete or stations missing
- No GNSS fix at the start
- Plots or sample trees fewer than planned
- Photos blurred or incomplete (method A or B)
- Measurements missing at some heights (method C)

D. Deviations from the protocol, problems and notes